Strategy-Space Modeling for Interpretable Quant Research

From Linear Combination to ML-Guided Allocation on GLD

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Abstract

This report presents a three-stage quantitative research framework applied to the GLD ETF. We treat trading strategy outputs as basis functions, constructing a design matrix whose columns represent strategy scores evaluated over time. Stage A evaluates single strategies in isolation. Stage B combines them via constrained optimization. Stage C applies supervised learning over an expanded 40-column basis to predict future returns and derive continuous portfolio weights. Stage C outperforms both prior stages in every year from 2020 to 2024, with 2024 returns of +16.4% (hold 130 days) and +25.2% (hold 45 days) at capital exposures of 65% and 78%, respectively.

1 Introduction and Motivation

This project originated from a mathematical observation made in class: polynomial regression uses a sequence of basis functions $1, x, x^2, \ldots, x^d$ to approximate an unknown relationship, and the key design decision is choosing which basis functions to include. The natural question that followed was:

Can trading strategies also be treated as basis functions?

If a trading strategy maps a market state on any given day to a numerical score, that score can be evaluated across all dates and placed into a design matrix. The columns become strategy functions, the rows become trading days, and the model selection problem becomes: which strategies, with which parameters, best predict future returns?

Table 1 formalizes this analogy.

Table 1: Function-space analogy between polynomial and strategy models.

Concept	Polynomial Model	Strategy Model
Basis function	$1,\ x,\ x^2,\ \ldots$	strategy signal functions
Data point	x_i	market state at date t_i
Matrix entry	$\phi_j(x_i)$	strategy score $s_j(t_i)$
Model selection	choose degree / penalty	choose strategy / penalty
Target	observed y_i	future cumulative return

The core design matrix is:

$$A_{ij} = \phi_j(\text{market state at } t_i)$$

where each column j is one strategy function evaluated over time, and each row i is one trading day.

2 Four Trading Strategy Families

We define four strategy families, each producing a daily numerical score that serves as one column of matrix A.

Table 2: Overview of the four strategy families.

Strategy	Signal Type	Core Idea
Adaptive Band	continuous	price position inside rolling bands
MA Crossover	continuous	short vs. long moving average gap
Adaptive Volatility Band	continuous	volatility regime from H-L-C range
Fear-Greed Candle-Volume	event	large candle + volume spike

Adaptive Band. The price is normalized relative to a rolling moving average. Upper and lower bands are placed at scaled thresholds. The score reflects how far the price sits from the band midpoint; buy signals occur near the lower band and sell signals near the upper band.

MA Crossover. A short-window moving average is compared against a long-window moving average. The score captures the direction and magnitude of the gap. A positive spread indicates an uptrend; a negative spread indicates a downtrend.

Adaptive Volatility Band. A volatility proxy derived from the high-low-close range is used to place adaptive bands. The score reflects the current volatility regime position. This strategy is qualitatively complementary to the price-band strategy.

Fear-Greed Candle-Volume. This event-style strategy fires only when a large candle body and a volume spike occur simultaneously. It captures extreme market sentiment and is rare by design. On GLD, after correcting a look-ahead bias, this strategy triggers near-zero exposure ($\approx 0\%$), which is an expected and correct result given GLD's typical low-volatility behavior.

2.1 Parameters as Signal Functions

Each strategy family has its own set of parameters. After optimization over a selection period, the best parameters define a concrete signal function for that anchor date. Parameters are re-optimized monthly using a grid search over each strategy's representative horizon (see Appendix A).

As an example, for anchor date 2021-12-31:

$$s_{\text{band}}(t) = \frac{P_t - \text{MA}_{65}(t)}{\sigma_{65}(t)}, \quad \text{buy when } s_{\text{band}}(t) < 0.6$$

$$s_{\text{ma}}(t) = \text{MA}_{60}(t) - \text{MA}_{65}(t), \quad \text{buy when } s_{\text{ma}}(t) > 0$$

Each anchor produces a different function adapted to recent market behavior, and the score value on any day becomes one entry in matrix A.

3 Stage A: Single Strategy Baseline

Stage A evaluates each strategy independently. Each strategy generates a binary signal (in or out of the market), and the portfolio holds for a fixed horizon of 130 days with monthly parameter updates.

Exposure in Stage A and B refers to the fraction of days the strategy holds a position: Exposure $= \frac{\#\text{ days in market}}{\text{total days}}$.

Table 3: Stage A annual returns and exposure. Hold 130 days, 1-month update.

Year	B&H	Adap. Band	MA Cross	Adap. Vol	Fear-Greed
2020	+23.9%	+4.4%/27%	+5.2%/29%	+5.8%/27%	$\approx 0\%$
2021	−6.2%	+0.3%/25%	−0.2%/21%	+0.9%/34%	$\approx 0\%$
2022	+0.8%	−0.9%/46%	−3.0%/18%	−2.1%/31%	$\approx 0\%$
2023	+11.8%	+1.5%/23%	+0.9%/25%	+2.1%/31%	$\approx 0\%$
2024	+27.0%	+5.9%/21%	+6.3%/32%	+10.6%/36%	$\approx 0\%$

Each strategy alone captures only a fraction of the market move. The low exposure (21–46%) reflects that the strategy sits out on most days. Fear-Greed produces near-zero returns on GLD due to infrequent triggering, which confirms correct behavior after removing a look-ahead bias in the original implementation.

4 Stage B: Ensemble via Constrained Optimization

Stage B asks whether combining the four strategy signals produces better results than any single signal.

4.1 Design

Define matrix $A \in \mathbb{R}^{n\times 4}$ where each column is one strategy score across n trading days. Find long-only weights $\mathbf{x} \geq 0$ that maximize realized portfolio return over the selection period:

$$\max_{\mathbf{x}\geq 0,\ \mathbf{1}^\top \mathbf{x}=1} \quad y(\mathbf{x}) = \text{portfolio return using combined signal } A\mathbf{x}$$

Weights are re-optimized at each anchor date and held fixed until the next anchor.

4.2 Optimizer Behavior

For anchor date 2021-12-31 (selection period return: +10.2% vs B&H −6.2%):

$$\hat{s}(t) = 0.45 \cdot s_{\text{band}}(t) + 0.00 \cdot s_{\text{ma}}(t) + 0.55 \cdot s_{\text{vol}}(t) + 0.00 \cdot s_{\text{fg}}(t)$$

The optimizer effectively performs Lasso-style selection: only the most informative signals (Adaptive Band and Adaptive Vol) receive nonzero weight, while MA Crossover and Fear-Greed are excluded.

4.3 Results

The ensemble beats the single strategy in 3 of 5 years, but the margin is small. In 2022, the ensemble performs *worse* (−2.3% vs −0.9%) because diversification across strategies amplifies losses

Table 4: Stage B ensemble results vs. Stage A. Hold 130 days, 1-month update.

Year	B&H	A: Best Single	B: Ensemble	Change
2020	+23.9%	+4.4%	+6.4%/32%	↑
2021	−6.2%	+0.3%	+0.3%/25%	∼
2022	+0.8%	−0.9%	−2.3%/42%	↓
2023	+11.8%	+1.6%	+2.0%/23%	↑
2024	+27.0%	+5.9%	+8.0%/29%	↑

in a whipsaw market. The fundamental limitation is that weights are fixed after optimization and cannot adapt to the current market regime.

5 Stage C: ML-Guided Allocation

Stage C reformulates the problem as supervised learning: rather than finding the best historical weight vector, we train a model to predict future returns from current strategy scores.

5.1 Expanding the Strategy Space

The design matrix is expanded from 4 to 40 columns by including the top-10 parameter sets per strategy family:

$$A \in \mathbb{R}^{n \times 40} \quad \text{(top 10 parameter sets} \times \text{4 strategy families)}$$

This mirrors the model-selection exercise from class: many basis functions are provided, and regularization selects the useful structure. Four linear model families are compared at each anchor:

$$\text{OLS} \quad \text{Ridge} \quad \text{Lasso} \quad \text{Elastic Net}$$

Model selection criterion: time-series cross-validated MSE (TimeSeriesSplit, no shuffling, 3–5 folds depending on selection period length).

5.2 Forward Validation Design

At each monthly anchor date, a 1-year selection period is used to fit all candidate models and select the best by CV-MSE. The chosen model is then applied to out-of-sample data in the evaluation period. The model is never exposed to future data during fitting:

$$\underbrace{\longleftarrow \text{1 year} \longrightarrow}_{\text{selection}} \Big|_{\text{anchor}} \underbrace{\longrightarrow}_{\text{evaluation}}$$

This walk-forward structure separates model discovery from validation.

5.3 From Prediction to Portfolio Weight

The selected model outputs a predicted future cumulative return $\hat{y}_t$ for each day. This is converted to a long-only portfolio weight:

$$w_t = \text{clip}\left(\frac{\hat{y}_t}{s},\ 0,\ 1\right)$$

where s is a quantile-based scale factor derived from the selection period.

Execution uses rolling tranches: each day a new tranche is opened at weight w_t and held for a fixed horizon (45 or 130 days). Gross exposure equals the average of all open tranches. When the model is confident ($\hat{y}_t$ large), exposure rises. When uncertain ($\hat{y}_t \approx 0$), exposure falls naturally.

5.4 What Exposure Means Across Stages

Stage A and B use a binary signal: each day is either fully in (100%) or out (0%). Their reported exposure is:

$$\text{Exposure}_{A,B} = \frac{\#\,\text{days in market}}{\text{total days}} \quad \text{(time-averaged binary)}$$

Stage C uses a true continuous daily capital weight $w_t \in [0, 1]$:

$$\text{Exposure}_C = \frac{1}{T}\sum_t w_t \quad \text{(true capital weight)}$$

This distinction matters for interpreting efficiency. Capital Efficiency is defined as Return ÷ Avg. Exposure, measuring return generated per unit of capital deployed.

Table 5: Capital efficiency comparison in 2024 (1-month update).

	Return	Avg. Exposure	Efficiency
Buy-and-Hold	+27.0%	100%	+27%
Stage C, hold 130 days	+16.4%	65%	+25%
Stage C, hold 45 days	+25.2%	78%	**+32%**

The hold-45 configuration generates a higher return per unit of capital deployed than buy-and-hold in 2024. Idle capital (22–35%) is available for deployment in other assets.

6 Results and Analysis

6.1 Three-Stage Comparison

Stage C outperforms Stage A and Stage B in every year. The progression in 2024 is particularly clear: A (+5.9%) → B (+8.0%) → C (+16.4%). The ML-guided approach learns not only which strategies to use but also *when* and *how much* to be exposed, which explains the consistent improvement over the fixed-weight Stage B.

Table 6: Three-stage comparison. Hold 130 days, 1-month update. Format: Return / Avg. Exposure.

Year	B&H	A: Single Signal	B: Ensemble	C: ML-Guided
2020	+23.9%	+4.4% / 27%	+6.4% / 32%	+**10.9**% / 57%
2021	−6.2%	+0.3% / 25%	+0.3% / 25%	+**0.8**% / 29%
2022	+0.8%	−0.9% / 46%	−2.3% / 42%	−**0.2**% / 31%
2023	+11.8%	+1.6% / 25%	+2.0% / 23%	+**4.0**% / 32%
2024	+27.0%	+5.9% / 21%	+8.0% / 29%	+**16.4**% / 65%

6.2 Update Frequency

Tables 7 and 8 compare model update intervals of 1, 2, 3, and 6 months for both hold horizons.

Table 7: Stage C update frequency comparison. Hold 130 days. Format: Return / Avg. Exposure.

Year	B&H	1M	2M	3M	6M
2020	+23.9%	+10.9% / 57%	+10.2% / 57%	+9.6% / 51%	+9.7% / 50%
2021	−6.2%	+0.8% / 29%	+0.3% / 35%	+0.6% / 28%	−0.7% / 49%
2022	+0.8%	−0.2% / 31%	+0.9% / 35%	−0.8% / 33%	−1.7% / 44%
2023	+11.8%	+4.0% / 32%	+4.1% / 23%	+3.8% / 40%	+3.4% / 15%
2024	+27.0%	+**16.4**% / 65%	+13.9% / 60%	+11.1% / 53%	+5.5% / 36%

Table 8: Stage C update frequency comparison. Hold 45 days. Format: Return / Avg. Exposure.

Year	B&H	1M	2M	3M	6M
2020	+23.9%	+17.4% / 72%	+16.5% / 71%	+13.7% / 64%	+13.3% / 63%
2021	−6.2%	+1.5% / 35%	+1.8% / 37%	+1.2% / 34%	−1.2% / 49%
2022	+0.8%	−0.6% / 37%	+1.3% / 42%	−0.6% / 38%	−2.5% / 51%
2023	+11.8%	+2.5% / 44%	+3.7% / 34%	+0.9% / 50%	+3.8% / 24%
2024	+27.0%	+**25.2**% / 78%	+19.8% / 73%	+16.2% / 66%	+11.0% / 52%

Monthly updates consistently outperform less frequent updates across both hold horizons. In 2024, the gap between 1M (+16.4%) and 6M (+5.5%) under h130 is substantial: faster adaptation captures regime shifts that slower updates miss.

6.3 Hold Horizon Comparison

Hold-45 captures bull markets more aggressively (2020, 2024) due to faster signal turnover. Hold-130 is more stable in flat or bearish conditions (2022, 2023). The h45 capital efficiency in 2024 (+32%) exceeds buy-and-hold (+27%), meaning the strategy generates more return per dollar deployed than a fully invested position.

Table 9: Hold horizon comparison. Stage C, 1-month update. Format: Return / Exposure / Efficiency.

Year	B&H	Hold 130 days	Hold 45 days
2020	+23.9% / +24% eff	+10.9% / 57% / +19%	+**17.4**% / 72% / +24%
2021	−6.2% / −6% eff	+**0.8**% / 29% / +3%	+1.5% / 35% / +4%
2022	+0.8% / +1% eff	−**0.2**% / 31% / −1%	−0.6% / 37% / −2%
2023	+11.8% / +12% eff	+**4.0**% / 32% / +12%	+2.5% / 44% / +6%
2024	+27.0% / +27% eff	+16.4% / 65% / +25%	+**25.2**% / 78% / +**32**%

7 Conclusion and Next Steps

7.1 Main Findings

Treating trading strategy outputs as basis functions creates a valid and predictive strategy space. The three-stage progression demonstrates clear and consistent improvement:

Table 10: Summary of findings across the three stages.

Stage	Key Takeaway
A: Single Signal	Exposure 21–46%; each strategy captures a small fraction of market move
B: Ensemble	Diversification alone does not reliably improve prediction
C: ML-Guided	Learns *when* to be exposed; 2024: ×2.0 vs B, ×2.8 vs A

The key advantage of Stage C is not merely higher returns but *dynamic exposure control*: capital is deployed when the model is confident and withheld when uncertain, leaving idle capital available for other assets.

7.2 Next Steps

1. **Multi-asset expansion**: apply the framework to BRK.B, oil, copper, and equity indices; deploy GLD idle capital into other assets when GLD exposure is low.

2. **Regime-aware exposure control**: add a second-layer signal that reduces global exposure during high-uncertainty market regimes.

3. **Nonlinear model selection via SINDy**: expand the library to $\Theta(A) = [s_i,\ s_i^2,\ s_i s_j,\ \ldots]$ and use sparse regression to select only the terms that matter.

4. **Hierarchical strategy structure**: direction signals feeding into timing signals, each layer applying a nonlinear transform.

7.3 Mini Project Team

Stages A and B were developed jointly with: Alejandro Menacho, Ashmeet Singh, Claire Wang, Thomas Tran, Matthew Aguirre, Sophia (Suyiao) Wei, Jack, and Stephen. Jack and Stephen additionally lead the live trading pipeline on a Raspberry Pi using Alpaca paper trading.

A Grid Search Parameter Ranges

Each strategy is optimized independently over its representative horizon.

Table 11: Grid search parameter ranges by strategy.

Strategy	Horizon	Parameter	Search Range
Adaptive Band	1y	`ma_window`	{5, 10, 22, 65, 130, 260} (horizon-filtered)
		`upper_k`	-3.0 to $+3.0$, step 0.1 (61 values)
		`lower_k`	-3.0 to $+3.0$, step 0.1 (61 values)
MA Crossover	6m	`short_ma`	5 to 200, step 5 (horizon-filtered)
		`long_ma`	5 to 200, step 5 (`short` $<$ `long`)
Adap. Vol. Band	3m	`vol_window`	{5, 10, 22, 65, 130, 260} (horizon-filtered)
		`upper_k`	-3.0 to $+3.0$, step 0.1
		`lower_k`	-3.0 to $+3.0$, step 0.1
Fear-Greed	1m	`k_body`	{1.2, 1.5, 1.8, 2.0, 2.5, 3.0}
		`k_volume`	{1.2, 1.5, 1.8, 2.0, 2.5, 3.0}
		`price_zone_window`	{3, 5, 7, 10}

Top-10 ranked parameter sets per strategy are passed to Stage C as basis columns (40 columns total). The body and volume baseline windows for Fear-Greed are fixed at 10 days and not optimized.